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Honors Chem

Heat Changes in Chemical Reactions

SCAN

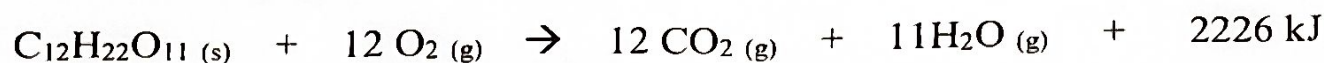
1. In the thermochemical equation for the decomposition of MgO into Mg and O, the ΔH_{rxn} is +1204 kJ/mol. Is the reaction exothermic or endothermic? What is the ΔH_{rxn} per mol of O₂? What is the ΔH_{rxn} per mole of MgO?



$$\frac{1204}{2 \text{ mol MgO}}$$

$$\frac{602}{1 \text{ mol O}_2}$$

2. In the thermochemical equation for the combustion of sucrose below, what is the ΔH_{rxn} /mole of sucrose? Is the reaction exothermic or endothermic? What is the ΔH_{rxn} in terms of H₂O?



it is exothermic

$$\frac{2226 \text{ kJ}}{1 \text{ mol}}$$

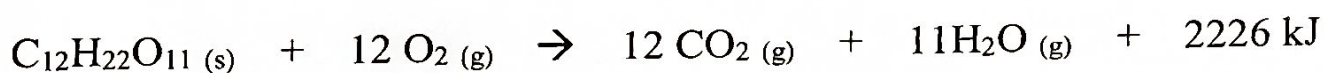
3. What is the ΔH if 15.0 g of hydrogen gas is converted to hydrogen fluoride in the following chemical reaction? Is the reaction exothermic or endothermic?



$$\Delta H_{\text{rxn}} = -536 \text{ kJ/mol}$$

$$\frac{-536 \text{ kJ}}{1 \text{ mol}} \times \frac{15.0 \text{ g}}{2.016 \text{ g/mol}} = -3990 \text{ kJ}$$

4. Refer to the combustion of sucrose reaction. If 171.15 g of sucrose is burned, what is the ΔH ? (note: the molar mass of sucrose is 342.30 g/mol). First write the ΔH_{rxn} in terms of sucrose.



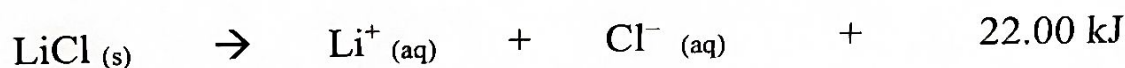
$$\frac{-2226 \text{ kJ}}{1 \text{ mol}} \times \frac{171.15 \text{ g}}{342.3 \text{ g/mol}} = -1113 \text{ kJ}$$

5. What is the ΔH for the production of NO₂ if 95.0 g of NO₂ are formed in the following equation?



$$\frac{-2857 \text{ kJ}}{2 \text{ mol}} \times \frac{95.0 \text{ g}}{46.005 \text{ g/mol}} = -2950 \text{ kJ}$$

6. What is the ΔH for the dissolution of 4.20 g of lithium chloride?



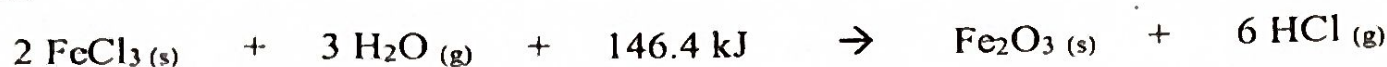
$$\frac{-22.00 \text{ kJ}}{1 \text{ mol}} \times \frac{4.20 \text{ g}}{42.394 \text{ g/mol}} = -2.18 \text{ kJ}$$

7. In the thermochemical equation below, write the ΔH_{rxn} /mole of Fe_2O_3 and in terms of Fe_3O_4 ? Is the reaction exothermic or endothermic, and how can you tell?



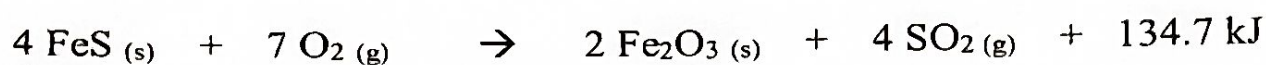
exo, heat on the right side
 $\frac{36.3 \text{ kJ}}{1 \text{ mol}} \quad -54\%$

8. What is the ΔH if 300.0 g of iron(III) chloride reacts with water to form iron(III) oxide and hydrochloric acid?



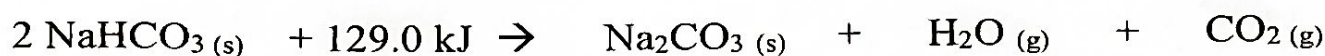
$$\frac{146.4 \text{ kJ}}{2 \text{ mol}} \times \frac{300.0 \text{ g}}{162.2 \text{ g/mol}} = 135.4 \text{ kJ}$$

9. In the reaction below, write the ΔH_{rxn} /mole of O_2 and per mole of Fe_2O_3 ? Is the reaction exothermic or endothermic, and how can you tell?



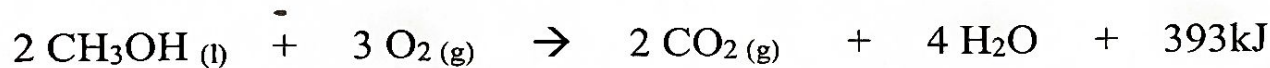
exothermic because it releases heat
 -10.24 kJ

10. What is the ΔH is for 1 tablespoon (14.00 g) of baking soda, NaHCO_3 , when it decomposes in the so-called "pancake reaction" to form light, fluffy and tasty pancakes.



$$\frac{129.0 \text{ kJ}}{2 \text{ mol}} \times \frac{14.00 \text{ g}}{84.01 \text{ g/mol}} = 10.75 \text{ kJ}$$

11. For the thermochemical equation for the combustion of methanol into carbon dioxide and water, What is the ΔH_{rxn} per mol of CH_3OH ? What is the ΔH_{rxn} per mole of O_2 ? Per mole of H_2O ?



$$\frac{-196.5 \text{ kJ}}{1 \text{ mol}} \quad \frac{-131 \text{ kJ}}{1 \text{ mol}} \quad \frac{-98.25 \text{ kJ}}{1 \text{ mol}}$$

12. What is something you are still confused about heat changes in chemical reactions?

Mixed-up answers:

-1113	-202.4	1204	-196.5	-2.18	602	-98.25	-36.3	-67.35
-3990	10.75	135.4	-131	-2950	-2226	-54.5	-19.24	